

A nonclosable Toeplitz matrix with arbitrary analytic part

Automated mathematical discovery run `fa_banach_001`, lane 11

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Status. Likely valid full counterexample to the existence question as printed.

Source and exact question

G. J. Groenewald, S. ter Horst, J. Jaftha, and A. C. M. Ran, “A Toeplitz-like operator with rational matrix symbol having poles on the unit circle: Matrix representation and spectral analysis,” arXiv:2409.19113, Remark 2.2. For $1 < p < \infty$, an operator $T : H_m^p \rightarrow H_m^p$ is in the Sarason class when it is closed and densely defined, its domain is shift invariant, and it satisfies the two additional shift conditions stated in the source. The authors ask:

Start with a Toeplitz matrix and assume the “analytic part” is in H_m^p . Can we then define an unbounded Toeplitz operator from that? It is clear how T acts on the vector polynomials of size m . Is there maybe a unique way to extend the operator on $\mathcal{P}^m$ to one that satisfies (i)–(iv)? Even without uniqueness this would be a relevant result.

that this extends to $1 < p < \infty$ (as already noted in [8]) and to arbitrary $m \in \mathbb{N}$ (for items (i)–(iii) see Proposition 2.1 in [9]).

Let us consider an operator T in the Sarason class with the property that the constant functions are in $\text{Dom}(T)$. Since $\text{Dom}(T)$ is T_{zI_m} -invariant by (ii), it follows that the vector polynomials of size m are in $\text{Dom}(T)$. Hence T admits a block matrix representation with respect to the standard block basis of H_m^p , and it is easy to see that (iii) implies that the block matrix representation has Toeplitz structure, i.e., $[T] = [a_{i-j}]_{i,j=0}^\infty$ for matrices $a_k \in \mathbb{C}^{m \times m}$.

In [8] we show that the operator in the Sarason class is not uniquely determined by the symbol, since for a scalar-valued rational function ω the adjoint T_ω^* of the unbounded Toeplitz operator T_ω is a restriction of the unbounded Toeplitz operator T_{ω^*} , and thus the symbol ω^* defines two different operators in the Sarason class, with the same symbol.

In short, for an operator in the Sarason class:

- (i) We get a matrix representation with respect to the standard basis, and this matrix representation has a Toeplitz structure.
- (ii) The Toeplitz matrix structure generates an abstract Fourier representation, and the fact that $\mathcal{P}^m \subset \text{Dom}(T)$ implies that the ‘analytic part’ must be in H_m^p . Apart from that, we are not sure if there are any conditions on the matrix entries that can be obtained.

From that perspective, we can state the following *open problem*. Start with a Toeplitz matrix and assume the ‘analytic part’ is in H_m^p . Can we then define an unbounded Toeplitz operator from that? It is clear how T acts on the vector polynomials of size m . Is there maybe a unique way to extend the operator on $\mathcal{P}^m$ to one that satisfies (i)–(iv)? Even without uniqueness this would be a relevant result.

Counterexample theorem

For matrices $a_j \in \mathbb{C}^{m \times m}$, the semi-infinite Toeplitz matrix is $[a_{i-j}]_{i,j \geq 0}$. Its forced action on a vector polynomial $f(z) = \sum_{j=0}^N x_j z^j$ is

$$T_0 f(z) = \sum_{i \geq 0} \left(\sum_{j=0}^N a_{i-j} x_j \right) z^i,$$

whenever the displayed output belongs to H_m^p .

Theorem 1. *Fix $1 < p < \infty$, $m \geq 1$, and any analytic matrix function*

$$G(z) = \sum_{j=0}^{\infty} A_j z^j \in H_{m \times m}^p.$$

Define a Toeplitz matrix by

$$a_j = A_j \quad (j \geq 0), \quad a_{-k} = k! I_m \quad (k \geq 1). \quad (1)$$

Its analytic part is $G \in H_{m \times m}^p$, and its matrix action maps $\mathcal{P}^m$ into H_m^p . Nevertheless, that polynomial operator is not closable. Consequently there is no closed operator on H_m^p extending the matrix action on $\mathcal{P}^m$, and in particular no extension in the Sarason class.

Proof intuition

The hypothesis controls only the nonnegative diagonals. Put factorial growth on the uncontrolled negative diagonals and test the matrix on a far-out monomial scaled by the same factorial. The inputs then vanish, but the top matrix entry survives as a nonzero constant; all other coanalytic entries and the prescribed analytic part disappear. This produces a nonvertical graph limit $(0, e)$ and destroys closability before any of the shift axioms can help.

Proof

Fix a nonzero $e \in \mathbb{C}^m$. Since $a_{i-j} = 0$ is not assumed for $i - j \geq 0$, we first separate the two parts. Applied to the monomial $z^n e$, the negative diagonals in (1) contribute only when $0 \leq i < n$, and the nonnegative diagonals contribute when $i \geq n$. Hence

$$T_0(z^n e) = \sum_{i=0}^{n-1} (n-i)! z^i e + z^n G(z) e. \quad (2)$$

This belongs to H_m^p ; by linearity the matrix therefore defines T_0 on every vector polynomial.

Set

$$f_n(z) = \frac{z^n}{n!} e.$$

Since $|z^n| = 1$ almost everywhere on the unit circle,

$$\|f_n\|_{H_m^p} = \frac{\|e\|}{n!} \longrightarrow 0. \quad (3)$$

After division of (2) by $n!$,

$$T_0 f_n = e + \sum_{i=1}^{n-1} \frac{(n-i)!}{n!} z^i e + \frac{z^n}{n!} G(z) e. \quad (4)$$

For $n \geq 3$, the middle term is uniformly bounded in norm by

$$\|e\| \sum_{i=1}^{n-1} \frac{(n-i)!}{n!} \leq \|e\| \left(\frac{1}{n} + \frac{2}{n(n-1)} \right) \rightarrow 0. \quad (5)$$

Indeed, after the first summand, each successive denominator gains a factor at least two. The last term in (4) has H_m^p norm

$$\frac{\|G(\cdot)e\|_{H_m^p}}{n!} \rightarrow 0. \quad (6)$$

Equations (4)–(6) give

$$T_0 f_n \rightarrow e \quad \text{in } H_m^p, \quad (7)$$

where e denotes the nonzero constant function.

Thus $f_n \rightarrow 0$ while $T_0 f_n \rightarrow e \neq 0$. This is the standard sequential criterion showing that T_0 is not closable. More directly, if a closed operator T extended T_0 , then closedness and (3), (7) would put $(0, e)$ in the graph of T , contradicting $T(0) = 0$. Since closedness is condition (i) of the Sarason class, no Sarason-class extension exists. $\square$

Verification and scope

The construction satisfies more than the minimal counterexample requires: it works for every $1 < p < \infty$, every block size m , and every prescribed admissible analytic part G . Thus adding or changing the controlled analytic diagonals cannot repair the obstruction. The theorem answers the unrestricted existence question negatively; it does not classify the additional growth or distribution conditions on the negative diagonals that would make the polynomial core closable or Sarason-extendable.

Novelty check

A bounded search on 17 August 2026 used the exact source question and title, as well as combinations of “Toeplitz matrix analytic part H^p ,” “Sarason class extension,” “closable,” and “factorial coanalytic coefficients.” It found the source, literature on rational symbols, semibounded Toeplitz forms, and general unbounded Toeplitz operators, but no answer to this exact matrix-extension question or this construction. The novelty assessment is “apparently new within the bounded search,” not an exhaustive bibliographic claim.

Human-review recommendation

Check the matrix indexing in (2) against the source convention $[a_{i-j}]_{i,j \geq 0}$. Once that is fixed, the uniform estimate and the closed-graph contradiction are elementary.

References

1. G. J. Groenewald, S. ter Horst, J. Jaftha, and A. C. M. Ran, “A Toeplitz-like operator with rational matrix symbol having poles on the unit circle: Matrix representation and spectral analysis,” arXiv:2409.19113.
2. D. Sarason, “Unbounded Toeplitz operators,” *Integral Equations Operator Theory* 61 (2008), 281–298.